CBSE Class 12 Chemistry — Amines

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

Q1. Arrange the following in the increasing order of their boiling points: $CH_3CH_2CH_2CH_2NH_2$, $(CH_3)_2NCH_2CH_3$, $CH_3CH_2NHCH_2CH_3$. (a) $(CH_3)_2NCH_2CH_3 < CH_3CH_2NHCH_2CH_3 < CH_3CH_2CH_2CH_2NH_2$ (b) $CH_3CH_2CH_2CH_2NH_2 < CH_3CH_2NHCH_2CH_3 < (CH_3)_2NCH_2CH_3$ (c) $(CH_3)_2NCH_2CH_3 < CH_3CH_2CH_2CH_2NH_2 < CH_3CH_2NHCH_2CH_3$ (d) $CH_3CH_2NHCH_2CH_3 < (CH_3)_2NCH_2CH_3 < CH_3CH_2CH_2CH_2NH_2$ [PYQ SQP 2022] [Confidence: Very High]

Q2. When ethanamide (CH_3CONH_2) reacts with $NaOH$ and Br_2 in an alcoholic medium, the major product formed is: (a) $CH_3CH_2NH_2$ (b) CH_3CH_2Br (c) CH_3NH_2 (d) CH_3COONa [PYQ 2020] [Confidence: Very High]

Q3. Which of the following amines will give a primary amine upon reduction with $LiAlH_4$? (a) CH_3CH_2NC (b) $CH_3CONHCH_3$ (c) CH_3CH_2CN (d) $(CH_3)_2CHNC$ [Practice] [Confidence: High]

Q4. The correct increasing order of basic strength of the following amines in the gaseous phase is: (a) $NH_3 < (C_2H_5)_3N < (C_2H_5)_2NH < C_2H_5NH_2$ (b) $NH_3 < C_2H_5NH_2 < (C_2H_5)_2NH < (C_2H_5)_3N$ (c) $NH_3 < C_2H_5NH_2 < (C_2H_5)_3N < (C_2H_5)_2NH$ (d) $(C_2H_5)_3N < (C_2H_5)_2NH < C_2H_5NH_2 < NH_3$ [PYQ 2019] [Confidence: Very High]

Q5. Which of the following reagents is used in the Gabriel phthalimide synthesis of primary amines? (a) Phthalimide, KOH, $R - X$ (b) Phthalic acid, NH_3, $R - X$ (c) Phthalimide, $NaOH$, $CHCl_3$ (d) Phthalic anhydride, H_2O, $R - X$ [Practice] [Confidence: High]

Q6. Organic compounds A, B, and C are amines having equivalent molecular weights. A and B on reaction with benzenesulphonyl chloride give white precipitates. The white precipitate obtained from compound B remains insoluble in $NaOH$. Compound B is a: (a) Primary amine (b) Secondary amine (c) Tertiary amine (d) Quaternary ammonium salt [PYQ SQP 2025-26] [Confidence: Very High]

Q7. Which of the following statements is NOT correct for amines? (a) Most alkyl amines are more basic than ammonia solution. (b) The pK_b value of ethylamine is lower than that of benzylamine. (c) CH_3NH_2 on reaction with nitrous acid releases NO_2 gas. (d) Hinsberg's reagent reacts with secondary amines to form sulphonamides. [PYQ SQP 2022] [Confidence: Very High]

Q8. The reagents used for the Sandmeyer reaction to convert a diazonium salt to chlorobenzene are: (a) Cu/HCl (b) Cu_2Cl_2/HCl (c) $Cl_2/FeCl_3$ (d) HBF_4/Δ [PYQ 2014] [Confidence: Very High]

Q9. Assertion (A): Tertiary amines are more basic than corresponding secondary and primary amines in the gaseous state. Reason (R): Tertiary amines have three alkyl groups which cause a strong +I (inductive) effect. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ SQP 2022] [Confidence: Very High]

Q10. Assertion (A): Aniline does not undergo Friedel-Crafts alkylation reaction. Reason (R): Aniline forms a complex with the Lewis acid $AlCl_3$, which acts as a catalyst in the Friedel-Crafts reaction. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2018] [Confidence: Very High]

Q11. Assertion (A): Primary amines have higher boiling points than tertiary amines of comparable molecular mass. Reason (R): Primary amines possess intermolecular hydrogen bonding while tertiary amines do not. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2016] [Confidence: Very High]

Q12. Write the IUPAC name of the compound: $CH_3 - NH - CH(CH_3)_2$. [PYQ 2017] [Confidence: High]

Q13. Provide a concise chemical reason: Why is aniline less basic than methylamine? [Practice] [Confidence: Very High]

Q14. What is the chemical name and formula of Hinsberg's reagent? [PYQ Term-2 2022] [Confidence: Very High]

Q15. Why is an aqueous solution of an aliphatic amine basic in nature? [Practice] [Confidence: High]

Q16. Write the IUPAC name of the following compound: $(CH_3)_2N - CH_2CH_3$. [PYQ 2017] [Confidence: High]

Q17. Assertion (A): Acetanilide gives a monosubstituted product with bromine water, unlike aniline which gives a trisubstituted product. Reason (R): The $-NHCOCH_3$ group heavily moderates the activating effect of the amino group due to resonance with the carbonyl group. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ Themes] [Confidence: Very High]

Q18. Name the specific reaction used to convert an amide into a primary amine containing one carbon atom less than the parent amide. [PYQ 2018] [Confidence: Very High]

Q19. Which has a higher pK_b value: aniline or ammonia? [Practice] [Confidence: Medium]

Q20. Methylamine in water reacts with ferric chloride solution to precipitate hydrated ferric oxide (ferric hydroxide). Why? [PYQ 2015] [Confidence: High]

2-Mark Questions (Short Answer)

Q21. Arrange the following compounds in the increasing order of their basic strength in an aqueous solution and briefly state the reason for this order: CH_3NH_2, $(CH_3)_3N$, $(CH_3)_2NH$ [PYQ Term-2 2022] [Confidence: Very High]

Q22. Give a simple chemical test with an observable outcome to distinguish between aniline and N-methylaniline. [PYQ 2020] [Confidence: Very High]

Q23. Give a simple chemical test with an observable outcome to distinguish between ethylamine and aniline. [PYQ 2018] [Confidence: Very High]

Q24. Account for the following: Ammonolysis of alkyl halides is generally not considered a good method to prepare pure primary amines. [CBSE SQP 2022] [Confidence: Very High]

Q25. Write the balanced chemical equations for the Carbylamine reaction. What is the specific observation in this test? [PYQ 2018] [Confidence: Very High]

Q26. Account for the following: Aniline is usually acetylated before carrying out its nitration reaction. [CBSE SQP 2022] [Confidence: Very High]

Q27. How will you seamlessly convert Nitrobenzene into aniline? Write the relevant chemical equation. [PYQ 2014] [Confidence: Very High]

Q28. Although the $-NH_2$ group in aniline is ortho/para directing, nitration of aniline with a nitrating mixture ($HNO_3 + H_2SO_4$) yields a substantial amount (about 47%) of m-nitroaniline. Explain why. [CBSE SQP 2022] [Confidence: Very High]

Q29. Outline the Gabriel phthalimide synthesis with the necessary chemical equations. Why cannot aromatic primary amines be prepared by this method? [PYQ Term-2 2022] [Confidence: Very High]

Q30. Write the structures of the main products A and B in the following sequence of reactions:
$C_6H_5N_2^+Cl^- \xrightarrow{CuCN/KCN} A \xrightarrow{H_2O/H^+} B$ [PYQ 2020] [Confidence: High]

Q31. Complete the following sequence of reactions by identifying the major products A and B:
$CH_3CH_2Cl \xrightarrow{NaCN} A \xrightarrow{H_2/Ni} B$ [PYQ 2015] [Confidence: High]

Q32. Account for the following observation: The pK_b of aniline is notably lower than the pK_b of m-nitroaniline. [CBSE SQP 2022] [Confidence: High]

3-Mark Questions (Short Answer)

Q33. Write the specific chemical equations required to bring about the following conversions: (i) Ethanoic acid to methanamine (ii) Aniline to p-bromoaniline (using protection) [PYQ 2014 / 2016] [Confidence: Very High]

Q34. Provide solid reasoning for the following statements: (i) Primary amines have significantly higher boiling points than tertiary amines. (ii) Aromatic diazonium salts are much more stable than aliphatic diazonium salts. (iii) $(CH_3)_2NH$ is more basic than $(CH_3)_3N$ in an aqueous solution. [PYQ 2014, 2018] [Confidence: Very High]

Q35. Describe the following important organic reactions with a suitable chemical equation for each: (i) Hoffmann bromamide degradation (ii) Sandmeyer reaction (iii) Coupling reaction of diazonium salts [PYQ 2018 / 2015] [Confidence: Very High]

Q36. Complete the following reactions by predicting the exact structures of the major products: (i) $C_6H_5NH_2 + CHCl_3 + KOH(alc) \xrightarrow{\Delta}$ (ii) $C_6H_5N_2^+Cl^- + H_3PO_2 + H_2O \rightarrow$ (iii) $C_6H_5NH_2 + Br_2(aq) \rightarrow$ [PYQ SQP Term-2 2022 / 2015] [Confidence: Very High]

Q37. Give reliable chemical tests to specifically distinguish between the following pairs of compounds: (i) Primary and secondary amines (ii) Aniline and benzylamine (iii) Methylamine and dimethylamine [PYQ 2019 / Practice] [Confidence: High]

Q38. An aromatic compound 'A' on treatment with aqueous ammonia and heating forms compound 'B', which on heating with Br_2 and KOH forms a compound 'C' having the molecular formula C_6H_7N. Identify the structures and write the IUPAC names of compounds A, B, and C. Write the chemical equations involved. [PYQ 2015] [Confidence: Very High]

Q39. How will you systematically carry out the following conversions? (i) Benzene to aniline (ii) Aniline to phenol (iii) Aniline to N-phenylethanamide [Practice / PYQ 2014] [Confidence: High]

Q40. Give chemical reasons for the following observations: (i) Aniline fails to undergo the Friedel-Crafts reaction. (ii) The acylation of aniline is typically carried out in the presence of a base like pyridine. (iii) Amines are considerably less acidic than alcohols of comparable molecular masses. [CBSE SQP Term-2 2022 / PYQ 2018] [Confidence: Very High]

Q41. Predict the structures of the main products in the following reactions: (i) $CH_3CONH_2 \xrightarrow{LiAlH_4/Ether}$ (ii) $C_6H_5N_2^+Cl^- \xrightarrow{CH_3CH_2OH}$ (iii) $CH_3CH_2Cl \xrightarrow{NH_3(excess),\Delta}$ [Practice] [Confidence: Medium]

Q42. Arrange the following compounds as directed and justify your answer: (i) Aniline, p-nitroaniline, and p-toluidine (Increasing order of basic strength) (ii) $C_6H_5NH_2$, $C_6H_5NHCH_3$, $C_6H_5CH_2NH_2$ (Increasing order of pK_b values) [Delhi Gov. SQP Term-2 2022 / PYQ 2018] [Confidence: High]

5-Mark Questions (Long Answer / Case-Study)

Q43. An organic compound A (C_3H_9N) reacts with benzenesulphonyl chloride to give a solid compound that is completely insoluble in alkali. (i) Deduce the exact structure of compound A and state whether it is a primary, secondary, or tertiary amine. (2 Marks) (ii) Write the balanced chemical equation for the reaction of A with benzenesulphonyl chloride. (1 Mark) (iii) What visual outcome would occur if an isomeric primary amine of A reacts with the same reagent? (1 Mark) (iv) Write the chemical equation for the carbylamine reaction of the primary amine isomer of A. (1 Mark) [Practice based on CBSE PYQ] [Confidence: Very High]

Q44. An organic compound 'A' with the molecular formula $C_7H_7NO_2$ exists in three isomeric forms. The isomer 'A' has the highest melting point of the three. 'A' on reduction gives compound 'B' with the molecular formula C_7H_9N. 'B' on treatment with $NaNO_2/HCl$ at 0–5°C forms compound 'C'. On treating 'C' with H_3PO_2, it gets converted to 'D' with formula C_7H_8, which on further reaction with CrO_2Cl_2 followed by hydrolysis forms 'E' (C_7H_6O). (i) Deduce the exact structures of compounds A, B, C, D, and E. (4 Marks) (ii) Write the specific chemical equation for the conversion of D to E and name the reaction. (1 Mark) [CBSE SQP 2023] [Confidence: Very High]

Q45. Case Study: Diazonium Salts Diazonium salts ($R - N_2^+X^-$) are highly versatile synthetic intermediates in organic chemistry. They are prepared by the reaction of primary amines with nitrous acid at low temperatures. While aliphatic diazonium salts are highly unstable and decompose to give alcohols and nitrogen gas, aromatic diazonium salts are relatively stable for a short time in cold aqueous solutions and can be used to introduce various functional groups onto the benzene ring. (i) Structurally, why are aromatic diazonium salts significantly more stable than aliphatic diazonium salts? (1 Mark) (ii) What is the major organic product formed when benzenediazonium chloride reacts with Copper cyanide ($CuCN$)? (1 Mark) (iii) Write the complete balanced chemical equation for the coupling reaction of benzenediazonium chloride with phenol. What is the colour of the resulting dye? (2 Marks) (iv) Name the specific reagent used to efficiently convert benzenediazonium chloride back to benzene. (1 Mark) [Practice / PYQ themes] [Confidence: Very High]

Q46. Account precisely for the following observations: (a) The diazotisation of aniline is strictly carried out in an ice-cold solution (0–5°C). (1 Mark) (b) Tertiary amines utterly fail to undergo acylation reactions with acid chlorides. (1 Mark) (c) When methylamine gas is bubbled into water and treated with an aqueous ferric chloride solution, a brown precipitate of ferric hydroxide is formed. (1 Mark) (d) The order of basicity of methylamines in an aqueous solution is strictly Secondary > Primary > Tertiary > Ammonia. (1 Mark) (e) Ethylamine freely dissolves in water, whereas aniline is completely insoluble. (1 Mark) [PYQ 2018, 2014, 2015] [Confidence: High]

Q47. Complete the following chemical reactions by identifying the major organic products formed: (i) $C_6H_5NH_2 + (CH_3CO)_2O \xrightarrow{\text{Pyridine}}$ (ii) $CH_3CH_2CN \xrightarrow{Na(Hg)/C_2H_5OH}$ (iii) $C_6H_5N_2^+Cl^- + HBF_4 \rightarrow \text{Intermediate} \xrightarrow{\Delta}$ (iv) $R - X + NH_3\ (\text{excess}) \xrightarrow{373K \text{ in sealed tube}}$ (v) $C_6H_5NO_2 \xrightarrow{Fe/HCl}$ (1 Mark each = 5 Marks) [Practice] [Confidence: Very High]

Q48. Write the complete chemical equations to carry out the following conversions logically: (i) Benzyl chloride to 2-phenylethanamine (ii) Chlorobenzene to p-chloroaniline (iii) Nitrobenzene to benzoic acid (iv) Aniline to benzyl alcohol (v) Ethanoic acid to methanamine (1 Mark each = 5 Marks) [PYQ Themes / Distinguish Tests] [Confidence: High]

Q49. Case Study: Reductive Alkylation of Amines Reductive alkylation is the term applied to the process of introducing alkyl groups into ammonia or a primary/secondary amine using an aldehyde or ketone in the presence of a reducing agent like hydrogen and a catalyst. The process consists of the initial addition of ammonia to a carbonyl compound to form an imine (Schiff's base), followed immediately by the reduction of this intermediate. (i) Ethanal on reaction with ammonia forms an imine (X), which on reaction with hydrogen gas and a metal catalyst gives (Y). Identify the exact structures of X and Y. (2 Marks) (ii) What specific type of reaction occurs between the initial ammonia and the aldehyde to form the imine? (1 Mark) (iii) Reductive alkylation of ammonia by means of an aldehyde using hydrogen as a reducing agent primarily results in the formation of which class of amines? (1 Mark) (iv) Why is reductive alkylation generally preferred over the direct ammonolysis of alkyl halides for the preparation of pure primary amines? (1 Mark) [CBSE Sample Paper Passage 2025-26 Theme] [Confidence: Very High]

Q50. A compound X with the molecular formula C_3H_7NO reacts aggressively with Br_2 in the presence of aqueous $NaOH$ to give a foul-smelling liquid compound Y. Compound Y reacts rapidly with HNO_2 (prepared in situ from $NaNO_2$ and HCl) at room temperature to form ethanol and rapid effervescence of N_2 gas. (i) Deduce the exact structures and IUPAC names of compounds X and Y. (2 Marks) (ii) Write the complete balanced chemical reaction for the conversion of X to Y. What is the name of this specific reaction? (2 Marks) (iii) What type of intermediate is formed in the reaction of Y with HNO_2 right before it releases N_2 gas? (1 Mark) [PYQ Reasoning Theme] [Confidence: High]

Answer Key

Q1. (a) $(CH_3)_2NCH_2CH_3 < CH_3CH_2NHCH_2CH_3 < CH_3CH_2CH_2CH_2NH_2$. (Tertiary < Secondary < Primary due to extent of intermolecular hydrogen bonding). **Q2.** (c) CH_3NH_2. (Hoffmann bromamide degradation reduces the carbon chain by one). **Q3.** (c) CH_3CH_2CN. (Nitriles reduce to primary amines; isocyanides reduce to secondary amines; amides reduce to corresponding amines). **Q4.** (b) $NH_3 < C_2H_5NH_2 < (C_2H_5)_2NH < (C_2H_5)_3N$. (In the gas phase, basicity strictly follows the +I effect of alkyl groups: $3^\circ > 2^\circ > 1^\circ > NH_3$). **Q5.** (a) Phthalimide, KOH, $R - X$. **Q6.** (b) Secondary amine. (Secondary amines form N,N-dialkylbenzenesulphonamide which lacks an acidic hydrogen, hence insoluble in NaOH). **Q7.** (c) CH_3NH_2 on reaction with nitrous acid releases NO_2 gas. (False, it releases N_2 gas). **Q8.** (b) Cu_2Cl_2/HCl. **Q9.** (a) Both A and R are true and R is the correct explanation of A. **Q10.** (a) Both A and R are true and R is the correct explanation of A. **Q11.** (a) Both A and R are true and R is the correct explanation of A. **Q12.** N-methylpropan-2-amine. **Q13.** In aniline, the lone pair of electrons on the nitrogen atom is delocalised over the benzene ring through resonance, making it less available for protonation compared to the highly localized lone pair in methylamine. **Q14.** Benzenesulphonyl chloride, $C_6H_5SO_2Cl$. **Q15.** Aliphatic amines react with water to accept a proton, forming an alkylammonium ion and releasing hydroxide ions (OH^-), which makes the solution basic. **Q16.** N,N-Dimethylethanamine. **Q17.** (a) Both A and R are true and R is the correct explanation of A. **Q18.** Hoffmann bromamide degradation reaction. **Q19.** Ammonia. (Aniline is a weaker base than ammonia, so its pK_b value is higher). **Q20.** Methylamine accepts a proton from water to yield OH^- ions ($CH_3NH_2 + H_2O \rightleftharpoons CH_3NH_3^+ + OH^-$). These OH^- ions react with Fe^{3+} ions to precipitate $Fe(OH)_3$.

Q21. $(CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$. This is due to the combined effects of the inductive (+I) effect, steric hindrance, and hydration (solvation) energy in the aqueous phase. **Q22.** Carbylamine test: Aniline (a 1° amine) when heated with $CHCl_3$ and alcoholic KOH gives a foul-smelling isocyanide. N-methylaniline (a 2° amine) does not give this test. **Q23.** Azo-dye test: Aniline on diazotization followed by coupling with alkaline 2-naphthol gives a brilliant orange-red dye. Ethylamine does not give this dye but releases brisk effervescence of N_2 gas. **Q24.** Ammonolysis yields a complex mixture of primary, secondary, and tertiary amines, and finally a quaternary ammonium salt, making it difficult to separate pure primary amines. **Q25.** $R - NH_2 + CHCl_3 + 3KOH \xrightarrow{\Delta} R - NC + 3KCl + 3H_2O$. Observation: Formation of an offensively foul-smelling gas (isocyanide). **Q26.** The $-NH_2$ group is highly activating and prone to oxidation by nitric acid. Acetylation protects the group, moderates its activating power, and prevents the formation of unwanted oxidation products and meta-derivatives. **Q27.** Reduction with Tin and Hydrochloric acid: $C_6H_5NO_2 + 6[H] \xrightarrow{Sn/HCl} C_6H_5NH_2 + 2H_2O$. **Q28.** In the strongly acidic nitrating mixture, a large portion of aniline is protonated to form the anilinium ion ($C_6H_5NH_3^+$), which is strongly deactivating and meta-directing. **Q29.** Phthalimide + $KOH \rightarrow$ Potassium phthalimide. Reacts with $R - X$ to give N-alkylphthalimide. Hydrolysis gives $R - NH_2$. Aromatic primary amines cannot be prepared because aryl halides do not undergo nucleophilic substitution with the phthalimide anion under normal conditions. **Q30.** A is Benzonitrile (C_6H_5CN). B is Benzoic acid (C_6H_5COOH). **Q31.** A is Propanenitrile (CH_3CH_2CN). B is Propan-1-amine ($CH_3CH_2CH_2NH_2$). **Q32.** Aniline is a stronger base than m-nitroaniline. The $-NO_2$ group exerts a strong electron-withdrawing effect (-I, -R), pulling electron density away from the ring and further reducing the availability of the nitrogen lone pair. Hence, aniline has a lower pK_b (higher basicity).

Q33. (i) $CH_3COOH \xrightarrow{NH_3,\Delta} CH_3CONH_2 \xrightarrow{Br_2/NaOH} CH_3NH_2$. (ii) $C_6H_5NH_2 \xrightarrow{(CH_3CO)_2O} C_6H_5NHCOCH_3 \xrightarrow{Br_2/CH_3COOH} p\text{-bromoacetanilide} \xrightarrow{H_2O/H^+} p\text{-bromoaniline}$. **Q34.** (i) Primary amines have two hydrogen atoms on nitrogen, leading to extensive intermolecular hydrogen bonding. Tertiary amines have no hydrogens on nitrogen. (ii) Aromatic diazonium salts are stabilized by resonance delocalization of the positive charge over the benzene ring. (iii) In $(CH_3)_2NH$, the combination of the +I effect and stabilizing hydration in water makes it more basic than $(CH_3)_3N$, which suffers from high steric hindrance to hydration. **Q35.** (i) $R - CONH_2 + Br_2 + 4NaOH \rightarrow R - NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$. (ii) $C_6H_5N_2^+Cl^- \xrightarrow{Cu_2Cl_2/HCl} C_6H_5Cl + N_2$. (iii) $C_6H_5N_2^+Cl^- + C_6H_5OH \xrightarrow{OH^-, pH 9-10} p\text{-hydroxyazobenzene (Orange dye)} + Cl^- + H_2O$. **Q36.** (i) C_6H_5NC (Phenyl isocyanide) + $3KCl + 3H_2O$. (ii) C_6H_6 (Benzene) + $H_3PO_3 + HCl + N_2$. (iii) 2,4,6-tribromoaniline (White ppt) + $3HBr$. **Q37.** (i) Hinsberg's reagent ($C_6H_5SO_2Cl$): 1° amine forms a precipitate soluble in NaOH. 2° amine forms a precipitate insoluble in NaOH. (ii) Azo dye test: Aniline forms an orange dye with $NaNO_2/HCl$ and 2-naphthol. Benzylamine gives no dye (releases N_2 gas). (iii) Carbylamine test: Methylamine gives a foul-smelling gas (CH_3NC). Dimethylamine does not. **Q38.** A is Benzoic acid (C_6H_5COOH). B is Benzamide ($C_6H_5CONH_2$). C is Aniline ($C_6H_5NH_2$). Equations: $C_6H_5COOH + NH_3 \xrightarrow{\Delta} C_6H_5CONH_2 + H_2O$. $C_6H_5CONH_2 + Br_2 + 4KOH \rightarrow C_6H_5NH_2 + K_2CO_3 + 2KBr + 2H_2O$. **Q39.** (i) Benzene $\xrightarrow{Conc.HNO_3 + Conc.H_2SO_4} C_6H_5NO_2 \xrightarrow{Sn/HCl} C_6H_5NH_2$. (ii) Aniline $\xrightarrow{NaNO_2/HCl, 0^\circ C} C_6H_5N_2Cl \xrightarrow{H_2O,\Delta} Phenol$. (iii) Aniline $\xrightarrow{(CH_3CO)_2O/\text{Pyridine}} C_6H_5NHCOCH_3$ (N-phenylethanamide). **Q40.** (i) The basic $-NH_2$ group forms a salt complex with the Lewis acid catalyst $AlCl_3$, completely deactivating the ring. (ii) Pyridine is added to neutralize the HCl formed during the reaction, driving the reaction forward. (iii) Oxygen is much more electronegative than Nitrogen, making the $O - H$ bond much easier to break to yield a proton compared to the $N - H$ bond. **Q41.** (i) $CH_3CH_2NH_2$ (Ethanamine). (ii) C_6H_6 (Benzene) + $CH_3CHO + N_2 + HCl$. (iii) $CH_3CH_2NH_2$ (Ethanamine as major product if NH_3 is in large excess). **Q42.** (i) p-nitroaniline < aniline < p-toluidine. (-NO2 withdraws electrons, decreasing basicity; -CH3 donates electrons, increasing basicity). (ii) $C_6H_5CH_2NH_2 < C_6H_5NHCH_3 < C_6H_5NH_2$. (Higher basicity means lower pKb. Aliphatic > 2° aromatic > 1° aromatic).

Q43. (i) Compound A is a secondary amine, N-methylethanamine ($CH_3 - NH - C_2H_5$). (ii) $CH_3 - NH - C_2H_5 + C_6H_5SO_2Cl \rightarrow C_6H_5SO_2N(CH_3)(C_2H_5) + HCl$. (iii) It would form a solid that dissolves completely in an aqueous alkali (NaOH) solution. (iv) $CH_3CH_2CH_2NH_2 + CHCl_3 + 3KOH \rightarrow CH_3CH_2CH_2NC + 3KCl + 3H_2O$. **Q44.** (i) A is p-nitrotoluene. B is p-toluidine. C is p-methylbenzenediazonium chloride. D is Toluene ($C_6H_5CH_3$). E is Benzaldehyde (C_6H_5CHO). (ii) $C_6H_5CH_3 + 2CrO_2Cl_2 \xrightarrow{CS_2} \text{Chromium complex} \xrightarrow{H_3O^+} C_6H_5CHO$. This is the Etard Reaction. **Q45.** (i) Aromatic diazonium salts are stabilized by the resonance delocalization of the positive charge of the diazonium group over the aromatic ring. Aliphatic diazonium salts lack this stabilization. (ii) Benzonitrile (C_6H_5CN). (iii) $C_6H_5N_2^+Cl^- + C_6H_5OH \xrightarrow{OH^-} C_6H_5 - N = N - C_6H_4OH$ (p-hydroxyazobenzene, Orange dye) $+HCl$. (iv) Hypophosphorous acid (H_3PO_2) with water, or Ethanol. **Q46.** (a) Diazonium salts are highly unstable at room temperature and decompose to give phenol. (b) Tertiary amines lack a replaceable hydrogen atom on the nitrogen. (c) Methylamine in water forms OH^- ions, which react with Fe^{3+} to precipitate $Fe(OH)_3$. (d) Due to the combined effects of inductive electron donation, steric hindrance, and solvation stabilizing factors in water. (e) Ethylamine forms strong hydrogen bonds with water. In aniline, the large bulky hydrocarbon (phenyl) part is hydrophobic and prevents solvation. **Q47.** (i) $C_6H_5NHCOCH_3$ (Acetanilide) + CH_3COOH. (ii) $CH_3CH_2CH_2NH_2$ (Propan-1-amine). (iii) C_6H_5F (Fluorobenzene) + $BF_3 + N_2$. (iv) $R - NH_2$ (Primary amine). (v) $C_6H_5NH_2$ (Aniline). **Q48.** (i) $C_6H_5CH_2Cl \xrightarrow{KCN} C_6H_5CH_2CN \xrightarrow{LiAlH_4} C_6H_5CH_2CH_2NH_2$. (ii) $C_6H_5Cl \xrightarrow{HNO_3/H_2SO_4} \textbf{p-nitrochlorobenzene} \xrightarrow{Sn/HCl} \textbf{p-chloroaniline}$. (iii) $C_6H_5NO_2 \xrightarrow{Sn/HCl} C_6H_5NH_2 \xrightarrow{NaNO_2/HCl} C_6H_5N_2Cl \xrightarrow{CuCN} C_6H_5CN \xrightarrow{H_3O^+} C_6H_5COOH$. (iv) $C_6H_5NH_2 \xrightarrow{NaNO_2/HCl} C_6H_5N_2Cl \xrightarrow{CuCN} C_6H_5CN \xrightarrow{H_3O^+} C_6H_5COOH \xrightarrow{LiAlH_4} C_6H_5CH_2OH$. (v) $CH_3COOH \xrightarrow{NH_3,\Delta} CH_3CONH_2 \xrightarrow{Br_2,NaOH} CH_3NH_2$. **Q49.** (i) X is $CH_3CH = NH$ (Ethanimine). Y is $CH_3CH_2NH_2$ (Ethanamine). (ii) Nucleophilic addition followed by elimination (condensation). (iii) Primary amines. (iv) Direct ammonolysis yields a mixed product of $1^\circ, 2^\circ, 3^\circ$ amines and quaternary salts, while reductive alkylation can specifically yield pure primary amines. **Q50.** (i) X is Propanamide ($CH_3CH_2CONH_2$). Y is Ethanamine ($CH_3CH_2NH_2$). (ii) $CH_3CH_2CONH_2 + Br_2 + 4NaOH \rightarrow CH_3CH_2NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$. Hoffmann bromamide degradation. (iii) An unstable aliphatic diazonium ion ($CH_3CH_2N_2^+$).
